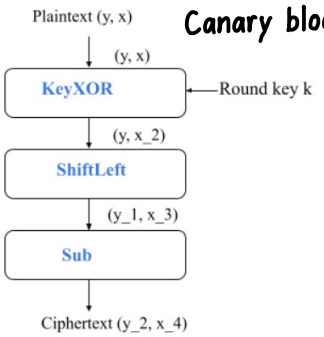
Braden Shiew Hao Wen 33273375 FIT2093 - Assignment 1

1.1 A)



Canary block cipher operations

Msg = 0xF2C5
key = 0xA5

Input : 16-bit plaintext block state (y,x) & round key k, where :
- x is the rightmost 8-bit of the block state
- y is the leftmost 8-bit of the block state
- k is an 8-bit round key

1st operation KeyXOR: ⊕ = bitwise exclusive or

$(y, x) \rightarrow (y, x_2)$

$x_2 = x \oplus k$

XOR Truth Table		
Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

First, convert the hexadecimal values into bits. Each value in binary is raised to the powers of 2, starting from the rightmost with 2^0. Each hex value will produce four binary digits.

x = C5 → C = 12 (8 4 2 1 = 1 1 0 0), 5 (8 4 2 1 = 0 1 0 1)

k = A5 → A = 10 (8 4 2 1 = 1 0 1 0), 5 (8 4 2 1 = 0 1 0 1)

$x \oplus k$ = 1100 0101 ⊕ 1010 0101

x_2 = 0110 0000 = 60_{hex}

2nd operation ShiftLeft:

$(y, x_2) \rightarrow (y_1, x_3)$

2-bit circular left-shift

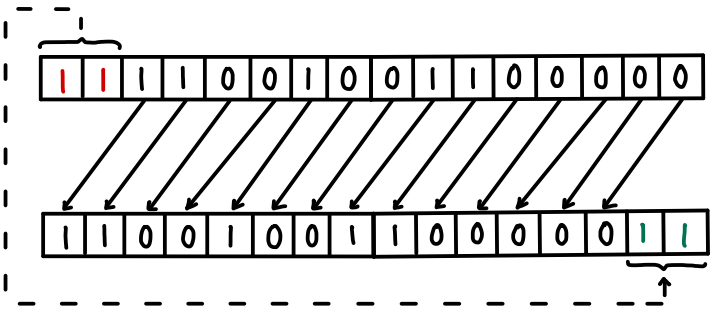
First, convert y hex into binary.

y = F2 → F = 15 (8 4 2 1 = 1 1 1 1), 2 (8 4 2 1 = 0 0 1 0)

Then, combine y and x_2 :

1111001001100000

Finally, perform 2-bit circular left-shift



$(y_1, x_3) = (11110010 \parallel 01100000) \ll 2$

$= 11001001\ 10000011$

$= (C9, 83)_{hex}$

3rd operation Sub:

$(y_1, x_3) \rightarrow (y_2, x_4)$

4-bit substitution on every 4-bit word of the state using the substitution table (S-box) for y_1 and x_3

$(y_1, x_3) = (C9, 83)$

$(y_2, x_4) = (C6, 49)$

Output: 16-bit ciphertext block state (y_2, x_4) where:
- rightmost 8-bit is x_4
- leftmost 8-bit is y_2

Canary S-box

Input (hex)	Output (hex)
0	7
1	F
2	0
3	9
4	3
5	1
6	D
7	B
8	4
9	6
A	2
B	8
C	C
D	E
E	5
F	A

1.1 B)

